

Walker Bush

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EDUCATION

Miami University

- Bachelor of Science
- Mechanical Engineering

Oxford, OH
Expected Graduation 2029
3.51 GPA

Saint Xavier High School

Honors & Awards

- National Honor Society
- KHSAA Academic All-State & Athletic All-State Team
- Alan Donhoff Award for Character & Leadership
- Tennis Coaches Award
- 2x Doubles State Runner Up Individual
- 2x Tennis Team Captain
- Principal's List (Junior & Senior Years)
- Trace of Hope Scholarship Recipient

Louisville, KY
Graduated 2025
3.9 GPA

RELEVANT EXPERIENCE

Kentuckiana Pool Management

Lifeguard

- Maintained a safe pool environment and responded to emergencies.
- Checked pool facilities and equipment to ensure safe operating conditions.

Louisville, KY
June 2023 - Present

Louisville Country Club

Tennis Instructor

- Led individual and group training sessions while adapting coaching to each player's needs.
- Built strong relationships with families and players to keep them engaged, motivated, and progressing throughout the season.

Louisville, KY
June 2025 - August 2025

Louisville Youth Advocacy Board

President

- Led initiatives to increase youth engagement and partnered with local organizations.
- Organized fundraisers that provided meals for children in need while mentoring board members.

Louisville, KY
August 2024 - May 2025

COURSES & PROJECT EXPERIENCE

Personal Project

Acebot QD001 + QD007

- Built and integrated an ESP32-based robotic car with 4-DOF robotic arm.
- Calibrated servo motors and configured code for full system operations.
- Diagnosed and resolved hardware, wiring, and software integration issues.

June – July 2026

CEC 112: Imagination Ingenuity Impact 2

Wind Turbine Project

- Designed, built, and tested a wind turbine using MATLAB and Arduino to collect and analyze performance data.
- Optimized blade design in gear issues through experimentation identifying the highest performing configuration.
- Collaborated on an HTML/CSS Website documenting the engineering process and results.

February – April 2026

CEC 111: Imagination Ingenuity Impact 1

Fluid Systems Design Project

- Redesigned a fluid system to improve reliability while reducing component count.
- Tested and refined multiple system iterations to improve performance.

October 2025

CEC 111: Imagination Ingenuity Impact 1

Electronic Monitoring System Project

- Built an Arduino based monitoring system and programmed it to detect fluid flow.
- Troubleshoot circuits and collaborated with teammates to improve reliability.

October 2025

ACTIVITIES

Kappa Alpha Order – Epsilon Lambda Chapter
Member

- Participated in chapter meetings, philanthropy events, and brotherhood activities.
- Worked with fellow members to support service projects, campus involvement, and professional development.

Oxford, OH
January 2026 - Present

Miami University Club Tennis Team
Member

- Competed in practices and matches while balancing academics at the collegiate club level.
- Collaborated with teammates to maintain a competitive and supportive team culture.

Oxford, OH
August 2025 - Present

Blessings in a Backpack
Mission Ambassador/Volunteer

- Assisted at frequent charity events, helping connect donors, volunteers, and community members to the cause.
- Led and coordinated fellow high school volunteers to support efficient food distribution and community outreach.

Louisville, KY
August 2019 - May 2025

Tiger Mentor Program
Mentor

- Mentored younger students by offering guidance on academics, athletics, and school involvement.
- Helped foster a welcoming and supportive school community.

Louisville, KY
August 2023 - May 2025

SKILLS / CERTIFICATIONS

Technical Skills	Software	Design Tools	Communication	Certifications
- Arduino Programming - Basic Circuit Analysis - Data Analysis - System Troubleshooting - Prototype Testing	- MATLAB - Claude Code - Arduino IDE - Microsoft Excel - Tableau - Figma - HTML/CSS	- Breadboards - Electronic Sensors - Servo Motor Calibration - Electronic Prototyping	- Team Collaboration - Public Speaking - Leadership - Project Coordination	- CPR/AED - First Aid - American Red Cross Lifeguard Certification